Double Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Iterated integrals over rectangles**

Evaluate $\int_0^2 \int_0^3 (x^2 + y)\, dy\, dx$.

Inner: $\int_0^3 (x^2 + y)\, dy = 3x^2 + \frac{9}{2}$. Outer: $\int_0^2 \left(3x^2 + \frac{9}{2}\right) dx = \left[x^3 + \frac{9}{2}x\right]_0^2 = 8 + 9 = 17$.

2 **[Foundational]** • **Target: Iterated integrals over rectangles**

Evaluate $\int_1^2 \int_0^1 6xy^2\, dx\, dy$.

Inner: $\int_0^1 6xy^2\, dx = 3y^2$. Outer: $\int_1^2 3y^2\, dy = [y^3]_1^2 = 8 - 1 = 7$.

3 **[Foundational]** • **Target: Double integrals over general regions**

Evaluate $\int_0^1 \int_0^x (x + y)\, dy\, dx$, integrating over the triangular region $0 \le y \le x \le 1$.

Inner: $\int_0^x (x + y)\, dy = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2}\, dx = \left[\frac{x^3}{2}\right]_0^1 = \frac{1}{2}$.

4 **[Foundational]** • **Target: Double integrals over general regions**

Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \le x \le 1$, $0 \le y \le 1$.

$\int_0^1 \int_0^1 (4 - x - y)\, dy\, dx$: inner $= 4 - x - \frac{1}{2}$. Outer: $\int_0^1 \left(\frac{7}{2} - x\right) dx = \frac{7}{2} - \frac{1}{2} = 3$.

5 **[Foundational]** • **Target: Reversing the order of integration**

Evaluate $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$ by first reversing the order of integration (the region is $0 \le y \le x \le 1$).

Reversing: $\int_0^1 \int_0^x e^{x^2}\, dy\, dx = \int_0^1 x e^{x^2}\, dx = \left[\frac{1}{2}e^{x^2}\right]_0^1 = \frac{e - 1}{2}$.

6 **[Foundational]** • **Target: Double integrals in polar coordinates**

Evaluate $\iint_D (x^2 + y^2)\, dA$ where D is the disk $x^2 + y^2 \le 4$, by converting to polar coordinates ($dA = r\, dr\, d\theta$).

$\int_0^{2\pi} \int_0^2 r^2 \cdot r\, dr\, d\theta = \int_0^{2\pi} \left[\frac{r^4}{4}\right]_0^2 d\theta = \int_0^{2\pi} 4\, d\theta = 8\pi$.

7 **[Foundational]** • **Target: Double integrals in polar coordinates**

Find the area of the region enclosed by $r = 2\cos\theta$ using a double integral in polar coordinates.

$A = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r\, dr\, d\theta = \int_{-\pi/2}^{\pi/2} 2\cos^2\theta\, d\theta = \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta)\, d\theta = \pi$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Average value over a region**

Find the average value of $f(x, y) = xy$ over the rectangle $0 \le x \le 2$, $0 \le y \le 3$ (average value $= \frac{1}{A}\iint_D f\,dA$, where A is the region's area).

$\iint_D xy\,dA = \int_0^2 \int_0^3 xy\,dy\,dx = \int_0^2 x \cdot \frac{9}{2}\,dx = \frac{9}{2} \cdot 2 = 9$. Area $= 6$. Average $= \frac{9}{6} = \frac{3}{2}$.

9 **[Intermediate]** • **Target: Mass and center of a region**

A lamina occupies the rectangle $0 \le x \le 2$, $0 \le y \le 1$ with density $\rho(x, y) = x + y$. Find its total mass, $M = \iint_D \rho\,dA$.

$M = \int_0^2 \int_0^1 (x + y)\,dy\,dx = \int_0^2 \left(x + \frac{1}{2}\right)dx = \left[\frac{x^2}{2} + \frac{x}{2}\right]_0^2 = 2 + 1 = 3$.

10 **[Intermediate]** • **Target: Mass and center of a region**

For a lamina occupying $0 \le x \le 1$, $0 \le y \le 1$ with constant density $\rho = 1$, find the x-coordinate of the center of mass, $\bar{x} = \frac{1}{M}\iint_D x\,dA$ (where M is the total mass).

$M = 1 \cdot 1 = 1$. $\iint_D x\,dA = \int_0^1 \int_0^1 x\,dy\,dx = \int_0^1 x\,dx = \frac{1}{2}$. $\bar{x} = \frac{1/2}{1} = \frac{1}{2}$ (as expected, the center of a uniform square).

11 **[Intermediate]** • **Target: Triple integral setup (a first look)**

Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \le x \le 1$, $0 \le y \le 2$, $0 \le z \le 3$.

$V = \int_0^1 \int_0^2 \int_0^3 dz\,dy\,dx$ (which evaluates to $1 \times 2 \times 3 = 6$, the box's volume).

12 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Evaluate $\int_0^1 \int_0^{y^2} 3x\,dx\,dy$, integrating over the region $0 \le x \le y^2$, $0 \le y \le 1$.

Inner: $\int_0^{y^2} 3x\,dx = \frac{3}{2}y^4$. Outer: $\int_0^1 \frac{3}{2}y^4\,dy = \left[\frac{3}{10}y^5\right]_0^1 = \frac{3}{10}$.

13 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Set up and evaluate the double integral for the volume under $f(x, y) = xy^2$ over the triangular region bounded by $y = x$, $y = 0$, and $x = 2$.

$V = \int_0^2 \int_0^x xy^2\,dy\,dx = \int_0^2 x \cdot \frac{x^3}{3}\,dx = \int_0^2 \frac{x^4}{3}\,dx = \left[\frac{x^5}{15}\right]_0^2 = \frac{32}{15}$.

14 **[Intermediate]** • **Target: An applied double-integral problem**

A metal plate occupies the region $0 \le x \le 3$, $0 \le y \le 2$, with temperature distribution $T(x, y) = 50 - x^2 - y^2$ (degrees Celsius). Find the average temperature over the plate.

$\iint_D T\,dA = \int_0^3 \int_0^2 (50 - x^2 - y^2)\,dy\,dx = \int_0^3 \left(100 - 2x^2 - \frac{8}{3}\right)dx = \int_0^3 \left(\frac{292}{3} - 2x^2\right)dx = \left[\frac{292}{3}x - \frac{2x^3}{3}\right]_0^3 = 292 - 18 = 274$. Area $= 6$. Average $= \frac{274}{6} \approx 45.67°\text{C}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Volume between two surfaces** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

Find the volume of the solid between $z = 8 - x^2 - y^2$ and $z = 0$ over the region where $z \geq 0$ requires $x^2 + y^2 \leq 8$, using polar coordinates.

$V = \int_0^{2\pi} \int_0^{2\sqrt{2}} (8 - r^2) r \, dr \, d\theta = \int_0^{2\pi} \left[4r^2 - \frac{r^4}{4}\right]_0^{2\sqrt{2}} d\theta = \int_0^{2\pi} (32 - 16) \, d\theta = \int_0^{2\pi} 16 \, d\theta = 32\pi.$

16 **[Challenging]** • **Target: Double integrals with exponential integrands** • **Cross-Lesson: 9.01 Separable Differential Equations**

Evaluate $\int_0^1 \int_0^1 x e^{xy} \, dy \, dx$ by integrating with respect to y first.

Inner: $\int_0^1 x e^{xy} \, dy = [e^{xy}]_0^1 = e^x - 1$ (for $x \neq 0$). Outer: $\int_0^1 (e^x - 1) \, dx = [e^x - x]_0^1 = (e - 1) - 1 = e - 2$.

17 **[Challenging]** • **Target: A capstone applied double-integral problem** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

An architect is designing a swimming pool whose floor depth varies according to $d(x, y) = 1 + 0.2x + 0.1y$ (meters) over a rectangular footprint $0 \leq x \leq 10$, $0 \leq y \leq 5$ (meters). Set up and evaluate the double integral for the total volume of water needed to fill the pool, then find the average depth by dividing by the pool's surface area.

$V = \int_0^{10} \int_0^5 (1 + 0.2x + 0.1y) \, dy \, dx$. Inner: $\int_0^5 (1 + 0.2x + 0.1y) \, dy = 5 + x + 1.25 = 6.25 + x$. Outer: $\int_0^{10} (6.25 + x) \, dx = \left[6.25x + \frac{x^2}{2}\right]_0^{10} = 62.5 + 50 = 112.5$ m^3. Surface area $= 10 \times 5 = 50$ m^2. Average depth $= \frac{112.5}{50} = 2.25$ m.

18 **[Challenging]** • **Target: A capstone applied double-integral problem** • **Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

For the double integral $\iint_D (2x + y) \, dA$ where D is the region bounded by $y = x^2$ and $y = x$, complete each step:

a Find the intersection points of the two curves to determine the bounds. $x^2 = x$ gives $x = 0, 1$. On $[0, 1]$, $x \geq x^2$, so $x^2 \leq y \leq x$.

b Set up the iterated integral with x as the outer variable. $\iint_D (2x + y) \, dA = \int_0^1 \int_{x^2}^{x} (2x + y) \, dy \, dx$.

c Evaluate the integral. Inner: $\left[2xy + \frac{y^2}{2}\right]_{x^2}^{x} = \left(2x^2 + \frac{x^2}{2}\right) - \left(2x^3 + \frac{x^4}{2}\right) = \frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}$. Outer: $\int_0^1 \left(\frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}\right) dx = \frac{5}{6} - \frac{1}{2} - \frac{1}{10} = \frac{25 - 15 - 3}{30} = \frac{7}{30}$.

(a) $x^2 = x$ gives $x = 0, 1$. On $[0, 1]$, $x \geq x^2$, so $x^2 \leq y \leq x$.; (b) $\iint_D (2x + y) \, dA = \int_0^1 \int_{x^2}^{x} (2x + y) \, dy \, dx$.; (c) Inner: $\left[2xy + \frac{y^2}{2}\right]_{x^2}^{x} = \left(2x^2 + \frac{x^2}{2}\right) - \left(2x^3 + \frac{x^4}{2}\right) = \frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}$. Outer: $\int_0^1 \left(\frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}\right) dx = \frac{5}{6} - \frac{1}{2} - \frac{1}{10} = \frac{25 - 15 - 3}{30} = \frac{7}{30}$.

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19 [Challenging] • Target: Reversing the order of integration • Cross-Lesson: 11.01 Sequences and Series

Evaluate $\int_0^{\sqrt{\pi}} \int_x^{\sqrt{\pi}} \sin(y^2)\,dy\,dx$ by reversing the order of integration.

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^{\sqrt{\pi}} \int_0^{y} \sin(y^2)\,dx\,dy$, which evaluates to 1.

20 [Challenging] • Target: Double integrals in polar coordinates • Cross-Lesson: 12.01 Convergence Tests

Use polar coordinates to evaluate $\iint_R e^{-x^2-y^2}\,dA$, where R is the disk $x^2+y^2 \le 9$.

With $x = r\cos\theta,\ y = r\sin\theta$ and $dA = r\,dr\,d\theta$, the integral becomes elementary; its value is $\frac{2^{\frac{189}{271}} \cdot 3^{\frac{128}{271}} \cdot 5^{\frac{275}{813}} \cdot 7^{\frac{97}{271}}}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Iterated Integrals over Rectangles

Evaluate $\int_0^2 \int_0^3 (xy^2)\,dx\,dy$.

a. 12 ✓ **b.** 18

c. 9 **d.** 6

Factor: $\left(\int_0^3 x\,dx\right)\left(\int_0^2 y^2\,dy\right) = [x^2/2]_0^3 [y^3/3]_0^2 = (9/2)(8/3) = 72/6 = 12$.

22 [Foundational] • Target: Iterated Integrals over Rectangles

Evaluate $\int_0^1 \int_0^2 (2x+3y)\,dy\,dx$.

a. 8 ✓ **b.** 6

c. 10 **d.** 4

Inner: $\int_0^2 (2x+3y)\,dy = [2xy + \tfrac{3}{2}y^2]_0^2 = 4x+6$. Outer: $\int_0^1 (4x+6)\,dx = [2x^2+6x]_0^1 = 2+6 = 8$.

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23 [Foundational] • Target: Iterated Integrals over Rectangles

What does Fubini's Theorem guarantee for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?

a. $\int_a^b \int_c^d f\,dy\,dx = \int_c^d \int_a^b f\,dx\,dy$ ✓

b. The integral equals the perimeter times height

c. The order of integration cannot be swapped

d. The integral must be zero

Fubini's Theorem ensures that the order of integration can be reversed without changing the value: $\iint_R f\,dA = \int_a^b \int_c^d f\,dy\,dx = \int_c^d \int_a^b f\,dx\,dy$.

24 [Foundational] • Target: Double Integrals over General Regions

Evaluate $\iint_D (x + y)\,dA$ where D is bounded by $y = 0$, $y = x$, and $x = 1$.

a. 1/2 ✓

b. 1/3

c. 2/3

d. 1/6

$\int_0^1 \int_0^x (x + y)\,dy\,dx = \int_0^1 [xy + y^2/2]_0^x\,dx = \int_0^1 (x^2 + x^2/2)\,dx = \int_0^1 \frac{3}{2}x^2\,dx = [\frac{x^3}{2}]_0^1 = 1/2.$

25 [Foundational] • Target: Double Integrals over General Regions

Reverse the order of integration for $\int_0^1 \int_x^1 f(x, y)\,dy\,dx$.

a. $\int_0^1 \int_0^y f(x, y)\,dx\,dy$ ✓

b. $\int_0^1 \int_y^1 f(x, y)\,dx\,dy$

c. $\int_0^1 \int_0^1 f(x, y)\,dx\,dy$

d. $\int_x^1 \int_0^1 f(x, y)\,dx\,dy$

The region is $0 \le x \le 1$ and $x \le y \le 1$. In terms of y: $0 \le y \le 1$, and for each y, x ranges from 0 to y. Thus $\int_0^1 \int_0^y f(x, y)\,dx\,dy$.

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26 [Foundational] • Target: Double Integrals over General Regions

Evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $\frac{e-1}{2}$ ✓ **b.** $e-1$

c. $\frac{e}{2}$ **d.** e^2-1

Reversed: $\int_0^1 \int_0^x e^{x^2}\,dy\,dx = \int_0^1 x e^{x^2}\,dx = [\frac{1}{2}e^{x^2}]_0^1 = \frac{e-1}{2}$.

27 [Foundational] • Target: Double Integrals in Polar Coordinates

What is the area element dA in polar coordinates?

a. $dA = r\,dr\,d\theta$ ✓ **b.** $dA = dr\,d\theta$

c. $dA = r^2\,dr\,d\theta$ **d.** $dA = \frac{1}{r}\,dr\,d\theta$

The Jacobian factor for polar coordinates is r, so the infinitesimal area element is $dA = r\,dr\,d\theta$.

28 [Foundational] • Target: Double Integrals in Polar Coordinates

Evaluate $\iint_R (x^2+y^2)\,dA$ over the disk $x^2+y^2 \le 4$.

a. 8π ✓ **b.** 16π

c. 4π **d.** 32π

In polar: $x^2+y^2 = r^2$. Bounds: $0 \le r \le 2, 0 \le \theta \le 2\pi$.
$\int_0^{2\pi} \int_0^2 r^2 (r\,dr)\,d\theta = 2\pi \int_0^2 r^3\,dr = 2\pi[r^4/4]_0^2 = 2\pi(4) = 8\pi$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Double Integrals in Polar Coordinates

Evaluate the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx$ using polar coordinates.

a. $\sqrt{\pi}$ ✓ **b.** π

c. $2\sqrt{\pi}$ **d.** $\pi/2$

$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA = \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-r^2} dr = 2\pi[-\frac{1}{2}e^{-r^2}]_0^{\infty} = 2\pi(1/2) = \pi \Rightarrow I = \sqrt{\pi}$.

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30 [Intermediate] • Target: Volume, Mass, and Center of Mass

Find the volume under the paraboloid $z = 4 - x^2 - y^2$ above the xy-plane.

a. 8π ✓

b. 16π

c. 4π

d. 2π

Region D: disk of radius $r = 2$. $V = \int_0^{2\pi}\int_0^2 (4 - r^2) r\,dr\,d\theta = 2\pi[2r^2 - r^4/4]_0^2 = 2\pi(8 - 4) = 8\pi$.

31 [Intermediate] • Target: Volume, Mass, and Center of Mass

A lamina occupies the region $[0, 1] \times [0, 2]$ with density $\rho(x, y) = 3x$. Find its total mass M.

a. 3 ✓

b. 6

c. 1.5

d. 4.5

$M = \int_0^1\int_0^2 3x\,dy\,dx = 2\int_0^1 3x\,dx = 2[\frac{3}{2}x^2]_0^1 = 3$.

32 [Intermediate] • Target: Volume, Mass, and Center of Mass

State the formula for the center of mass $\bar{x}$ of a lamina with density $\rho(x, y)$ and total mass M.

a. $\bar{x} = \frac{1}{M}\iint_D x\rho(x, y)\,dA$ ✓

b. $\bar{x} = \frac{1}{M}\iint_D y\rho(x, y)\,dA$

c. $\bar{x} = \iint_D x\,dA$

d. $\bar{x} = \frac{M}{\iint x\,dA}$

The x-coordinate of the center of mass is the moment about the y-axis divided by total mass: $\bar{x} = \frac{M_y}{M} = \frac{1}{M}\iint_D x\rho(x, y)\,dA$.

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33 [Intermediate] • Target: Iterated Integrals over Rectangles

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.

a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$ ✓

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_0^4((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_1^4((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as r. The volume of the material removed by the cylindrical.

34 [Intermediate] • Target: Double Integrals over General Regions

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$ ✓

c. 16π

d. $\frac{64\pi}{3}$

The solid E is bounded below by the paraboloid $z = x^2 + y^2$ and above by the paraboloid $z = 8 - x^2 - y^2$. To find the intersection of these two surfaces, we set them equal: $x^2 + y^2 = 8 - x^2 - y^2$, which simplifies to $2(x^2 + y^2) = 8$, or $x^2 + y^2 = 4$. This is a circle of radius 2 in the xy-plane. In cylindrical coordinates, $x = r\cos(\theta)$, $y = r\sin(\theta)$, and $z = z$. The equations of the surfaces become $z = r^2$ and $z = 8 - r^2$. The intersection occurs at $r^2 = 8 - r^2$, so.

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35 **[Intermediate]** • **Target: Double Integrals in Polar Coordinates**

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

36 **[Intermediate]** • **Target: Volume, Mass, and Center of Mass**

Iterated integral over tetrahedron $x + y + z \le 1, x, y, z \ge 0$: volume = $1/6$. The integral $\iiint x\, dV =$

a. $1/6$

b. $1/24$ ✓

c. $1/12$

d. $1/8$

By symmetry $\iiint x = \iiint y = \iiint z$; and $\iiint (x + y + z) = 3\iiint x$; by integration $\iiint (x + y + z)dV = 1/8$... actually $\iiint x\, dV = 1/24$ (standard result).

37 **[Intermediate]** • **Target: Iterated integrals over rectangles**

The region between $y = 2x$ and $y = x^3$ in the first quadrant is revolved about the x-axis. Find the volume.

a. $\frac{64\sqrt{2}\pi}{105}$

b. $\frac{16\sqrt{2}\pi}{15}$

c. $\frac{32\sqrt{2}}{21}$

d. $\frac{32\sqrt{2}\pi}{21}$ ✓

The curves meet at $x = 0$ and $x = \sqrt{2}$. Washers: $V = \pi\int_{0}^{\sqrt{2}} (-x^6 + 4x^2)\, dx \;=\; \frac{32\sqrt{2}\pi}{21}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Double integrals over general regions • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

The region between $y = 2x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{32\sqrt{2}\pi}{21}$

b. $\frac{8\sqrt{2}\pi}{15}$

c. $\frac{16\sqrt{2}}{15}$

d. $\frac{16\sqrt{2}\pi}{15}$ ✓

The curves meet at $x = 0$ and $x = \sqrt{2}$. Shells: $V = 2\pi\int_0^{\sqrt{2}} x(-x^3 + 2x)\,dx = \frac{16\sqrt{2}\pi}{15}$.

39 [Challenging] • Target: Reversing the order of integration • Cross-Lesson: 11.01 Sequences and Series

The region between $y = 3x$ and $y = x^2$ in the first quadrant is revolved about the x-axis. Find the volume.

a. $\frac{162\pi}{5}$ ✓

b. $\frac{81\pi}{10}$

c. $\frac{27\pi}{2}$

d. $\frac{162}{5}$

The curves meet at $x = 0$ and $x = 3$. Washers: $V = \pi\int_0^{3} (-x^4 + 9x^2)\,dx = \frac{162\pi}{5}$.

40 [Challenging] • Target: Double integrals in polar coordinates • Cross-Lesson: 12.01 Convergence Tests

The region between $y = 2x$ and $y = x^2$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{8\pi}{3}$ ✓

b. $\frac{64\pi}{15}$

c. $\frac{4\pi}{3}$

d. $\frac{8}{3}$

The curves meet at $x = 0$ and $x = 2$. Shells: $V = 2\pi\int_0^{2} x(-x^2 + 2x)\,dx = \frac{8\pi}{3}$.

41 [Challenging] • Target: Average value over a region • Cross-Lesson: 12.02 Power Series and Taylor Series

The region between $y = x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{4\pi}{21}$

b. $\frac{4\pi}{15}$ ✓

c. $\frac{2\pi}{15}$

d. $\frac{4}{15}$

The curves meet at $x = 0$ and $x = 1$. Shells: $V = 2\pi\int_0^{1} x(-x^3 + x)\,dx = \frac{4\pi}{15}$.

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42 [Challenging] • Target: Mass and center of a region • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The double integral over region D of $f = 1$ gives:

a. area of D ✓

b. perimeter of D

c. volume

d. 0

Applying Multiple Integrals principles, $\iint_D 1\,dA = \text{Area}(D)$. Therefore, the verified correct option is area of D.

43 [Challenging] • Target: Triple integral setup (a first look) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

The region between $y = x$ and $y = x^2$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{\pi}{6}$ ✓

b. $\frac{2\pi}{15}$

c. $\frac{\pi}{12}$

d. $\frac{1}{6}$

The curves meet at $x = 0$ and $x = 1$. Shells: $V = 2\pi\int_0^1 x(-x^2 + x)\,dx = \frac{\pi}{6}$.

44 [Challenging] • Target: Double integrals with a variable upper limit • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_a^b (5 - x)\sqrt{x}\,dx$. Find the value of k.

a. 2

b. 4/3

c. 8/3 ✓

d. 16/3

The region R is bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. We are revolving this region about the vertical line $x = 5$. We will use the method of cylindrical shells. The radius of a cylindrical shell at a given x is the distance from the axis of revolution $x = 5$ to x, which is $5 - x$. The height of the shell is the difference in the y-values of the bounding curves, which is $\sqrt{x} - 0 = \sqrt{x}$. The thickness of the shell is dx. The volume of a single cylindrical shell is.

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45 [Challenging] • Target: An applied double-integral problem • Cross-Lesson: 2.01 Evaluating Limits

Evaluate $\int_0^1 \int_{\sqrt{y}}^1 \sqrt{x^3+1}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{2}{9}+\frac{4\sqrt{2}}{9}$ ✓

b. $-\frac{4}{9}+\frac{8\sqrt{2}}{9}$

c. $\frac{4\sqrt{2}}{9}+\frac{7}{9}$

d. $-\frac{1}{9}+\frac{2\sqrt{2}}{9}$

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^1 \int_0^{x^2} \sqrt{x^3+1}\,dy\,dx$, which evaluates to $-\frac{2}{9}+\frac{4\sqrt{2}}{9}$.

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